Protein binding sites: 6 Å vs 12 Å

Total, 2-body (2B) and 2-body + 3-body (2B+3B) interaction energies for the four active sites, kcal/mol. DFT is the legacy ωB97XD reference from cafquant `mbe/RESULTS.md`. Models: plain charge/spin MACE (30 epochs) and from-scratch PolarMACE, each at r_max 6 and 12 Å, trained on the same 20k OMol25 frames.

Total

Table 1

System	DFT	MACE 6 Å	MACE 12 Å	POLAR 6 Å	POLAR 12 Å
sult	−109.5	−100.1	+175.7	−90.0	−209.1
ddc	−77.7	−174.9	−670.6	−267.4	−508.4
comt	−172.1	−261.6	−266.5	−301.2	−408.2
aldh	−50.4	−64.6	+233.2	−60.5	−174.5

Error vs DFT (model − DFT); the smaller of each 6/12 pair in bold:

Table 2

System	MACE 6 Å	MACE 12 Å	POLAR 6 Å	POLAR 12 Å
sult	**+9.4**	+285.2	**+19.5**	−99.6
ddc	**−97.2**	−592.9	**−189.6**	−430.7
comt	**−89.5**	−94.4	**−129.1**	−236.1
aldh	**−14.3**	+283.6	**−10.1**	−124.1

2B

Table 3

System	DFT	MACE 6 Å	MACE 12 Å	POLAR 6 Å	POLAR 12 Å
sult	−121.4	+87.2	−112.4	−101.9	−194.4
ddc	−83.1	−170.0	−401.0	−431.2	−584.2
comt	−290.7	−223.4	−286.0	−490.9	−644.9
aldh	−51.1	−79.2	−140.8	−48.4	−154.2

Error vs DFT (model − DFT); the smaller of each 6/12 pair in bold:

Table 4

System	MACE 6 Å	MACE 12 Å	POLAR 6 Å	POLAR 12 Å
sult	+208.6	**+9.0**	**+19.5**	−73.0
ddc	**−86.9**	−317.9	**−348.2**	−501.1

comt	+67.3	**+4.7**	**−200.2**	−354.2
aldh	**−28.1**	−89.8	**+2.7**	−103.1

2B+3B

Table 5

System	DFT	MACE 6 Å	MACE 12 Å	POLAR 6 Å	POLAR 12 Å
sult	−118.4	−199.9	−806.3	+102.9	−18.5
ddc	−194.9	−973.9	−1132.2	+347.2	+86.7
comt	+218.2	+1132.6	−2536.1	+104.2	+543.4
aldh	—	−65.9	+246.4	−80.2	−227.0

Error vs DFT (model − DFT); the smaller of each 6/12 pair in bold:

Table 6

System	MACE 6 Å	MACE 12 Å	POLAR 6 Å	POLAR 12 Å
sult	**−81.4**	−687.9	+221.3	**+100.0**
ddc	**−779.0**	−937.4	+542.0	**+281.6**
comt	**+914.3**	−2754.4	**−114.0**	+325.1

2B+3B does not converge for the charged systems even at the DFT level (cafquant `mbe/RESULTS.md`); compare it with care. Per-frame energies: `out/<system>/sp_lr-*.json`; full table for every backend: `table_all.txt`.